Statistical Foundations for Reliable AI Reasoning under Compute Constraints

Brief topic description for Program Officer discussion — ONR FY27 YIP (N0001426SF004)

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Proposed ONR S&T Focus Area: Artificial Intelligence and Autonomy — Machine Learning

Problem. Large language models can spend more computation on difficult questions by sampling reasoning paths, retrieving additional evidence, and asking process reward models (PRMs) to score intermediate steps. Yet more computation does not always help: ten samples from the same model may share the same faulty premise. For example, an agent answering whether a port is reachable may generate twenty plans, all based on one outdated tide report. A majority vote then gives false confidence while using twenty times the inference budget. Fixed best-of-N and voting schemes do not quantify the information gained from another sample.

Central hypothesis. A statistical model that tracks dependence among sampled reasoning paths and the uncertainty of their verifiers can allocate inference-time computation more efficiently while controlling errors among accepted answers. Unlike peer panels designed to influence human trust, the object here is a single system's internal sequential sampling and stopping policy.

Research plan.

- **Estimate the information in dependent traces.** Model a reasoning path as linked assertions with latent validity and noisy verifier observations. Paths can share prefixes, retrieved documents, and error modes. We will develop hierarchical or latent-variable estimators of effective independent evidence, study what can be identified from a small labeled calibration set and many unlabeled traces, and quantify uncertainty when error correlations cannot be learned.
- **Decide when to compute and when to stop.** Cast sampling, checking a step, retrieving another source, and stopping or abstaining as a sequential decision problem. We will investigate value-of-information policies with a target selective error rate and limits on expected tokens, retrieval calls, and latency. Distribution-free risk control and sequential testing will guide what guarantees survive dependent samples or task shift.
- **Make the verifiers useful for decision-making.** Train complementary process verifiers using synthetic factual and logical errors, with separate measurements of miscalibration and shared failures. Instead of blindly choosing the highest score, the policy will distinguish a genuinely independent check from redundant re-scoring. We will examine failure regimes in which further sampling increases cost but cannot reduce risk.

Evaluation and expected outcomes. MEQA (NeurIPS 2024), MuSiQue, and MATH will test multihop and quantitative reasoning; synthetic shared-prefix and shared-source perturbations will reveal known dependencies. We will report risk–coverage and accuracy–compute curves, calibration, and error control under held-out domains. Baselines include self-consistency and fixed-budget PRM best-of-N search.

Naval relevance. On a bandwidth-limited shipboard planning assistant, a question about safe passage may be answerable after one verified check or may need escalation because every candidate draws from the same stale source. The system will decide whether another local computation or external evidence request is worth its power, time, and connectivity cost. This addresses Mathematical Data Science priorities in principled information integration and the inference–computation trade-off.

Prior work and PI qualifications. Our synthetic-error verifier FG-PRM (Li et al., 2025), peer-judgment framework PRD (Li et al., 2024), and hierarchical process rewards for retrieval, HiPRAG (Wu et al., 2026), provide the starting point. The PI holds a 2024 NSF CAREER award and led RESIN's DARPA KAIROS delivery-system integration as a postdoc; the team achieved top performance among participants.